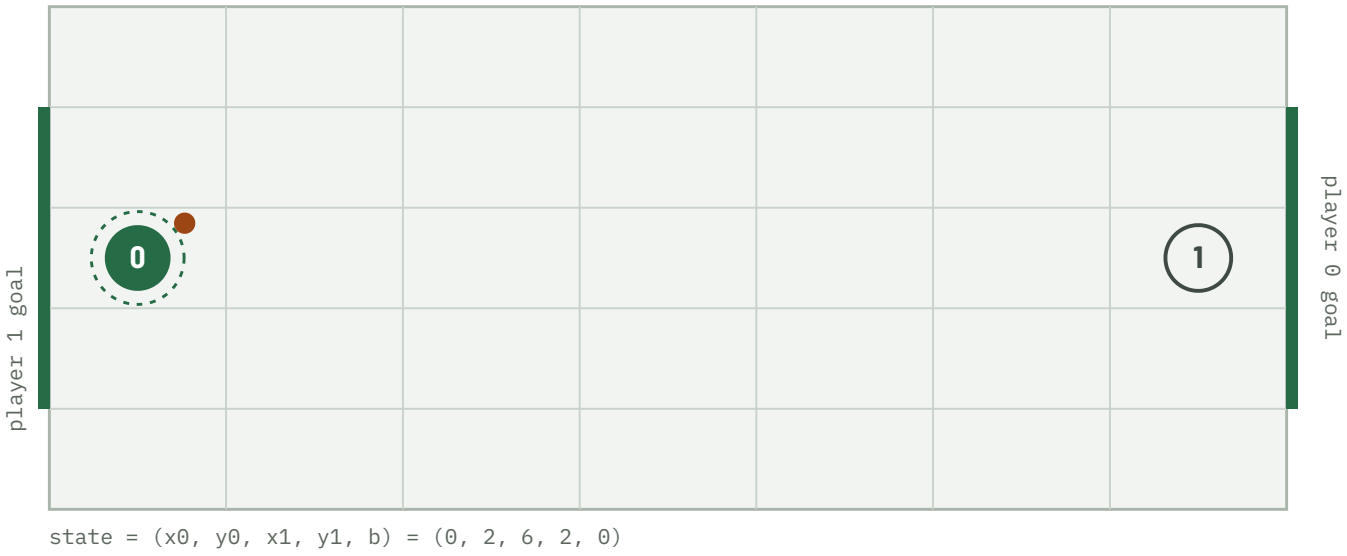
Pure-First Nash Q for Soccer

Where a two-player soccer Markov game does — and does not — need mixed strategies, and how much matrix-game work that saves.

FOUR QUESTIONS

RQ1 Existence — under what transition structures does a pure equilibrium exist at every state? **RQ2 Efficiency** — how much work does checking for a pure saddle first eliminate? **RQ3 Mechanism** — which spatial configurations force mixing? **RQ4 Robustness** — how do discounting and tolerance affect the split?

repo · soccer-markov-nash | 150+ tests | tables from experiments/*.csv | scope in docs/assumptions.md



The kickoff. 7×5 grid, 2 380 non-terminal states. Player 0 carries the ball and attacks the right edge; player 1 the left. Actions U D L R are chosen simultaneously; reward is +1 / −1 / 0, zero-sum.

ABSTRACT

1 **For the implemented deterministic A10 model, every one of the 2380 converged stage games has a pure saddle.** Playing a pure saddle action everywhere is then a stationary pure-strategy equilibrium, found in 9 sweeps with no linear program.

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2 **A coin-flip tie-break does not change this;** Littman's random *move order* does — but only when the goal mouth is more than one cell wide, and even then on ~4% of states (this configuration).

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3 **Those mixed states are geometrically localised:** a decision tree predicts them from the state with precision 0.96, and 68 of 94 reduce to a 2×2 matching-pennies support.

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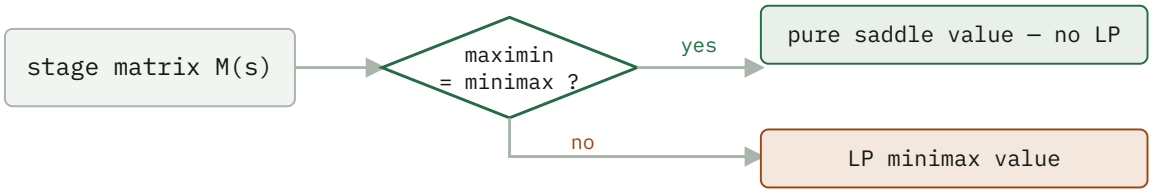
4 **The pure-first hybrid solver** matches the all-LP value function to 4e−16 while doing ~25× less matrix-game work; mirror symmetry halves it again.

Method

The game is zero-sum, so at every state the stage game is a payoff matrix $M(s)[a_0, a_1] = E[R + \gamma V(s')]$ and the Shapley (1953) fixed point is the minimax value:

$$V(s) = \mathrm{val}(M(s)) = \max_p \min_{k^3} (p^\top M)[a_1] = \min_p \max_{a0} (Mq)[a_0]$$

The $Q = R + \beta\,\mathrm{Nash}(Q')$ update is this operator. Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via `scipy` HiGHS), and **hybrid** (the pure saddle value where `maximin = minimax`, the LP elsewhere — exact everywhere). Support enumeration (`support_enum.py`) finds *all* equilibria, including of general-sum games.



The pure-first hybrid. On the deterministic game the "yes" branch is taken at every one of the 2 380 states, so the LP is never called.

RQ1 — Existence

MOVE ORDER	STOCHASTIC	CONVERGED STAGE GAMES WITH NO PURE SADDLE	STATIONARY PURE EQ?
deterministic (exact A10)	no	0 / 2380	yes (Claim B)
coinflip tie-break	yes	0 / 2380	yes (Claim B)
random move order	yes	94 / 2380 (this config)	no

IT IS THE MOVE ORDER, NOT THE RANDOMNESS

The deterministic and coin-flip value functions are *identical* (`max |V_det − V_coin| = 0` at every γ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames.

RQ2 — Efficiency

RANDOM MOVE ORDER, Γ = 0.9 — THE CASE WHERE THE LP IS ACTUALLY NEEDED

SOLVER	SWEEPS	V(KICKOFF)	MATRIX-GAME SOLVES	WALL CLOCK
pure (lower bound)	18	+0.000	0	<1 s
hybrid exact	108	+0.150	~9 700	~14 s
mixed exact	108	+0.150	~3.6 M	~8 min

The hybrid solves an LP on only the ~4% of states lacking a pure saddle — ~25× fewer — and matches the all-LP value to 4e−16. **Mirror symmetry** (`run_symmetric`): the game is anti-symmetric under board-flip + player-swap and every state has a distinct mirror, so the 2 380 states are 1 190 pairs — solve one, reconstruct the other by `V(mirror(s)) = −V(s)`.

Freeze-then-iterate, and its staleness

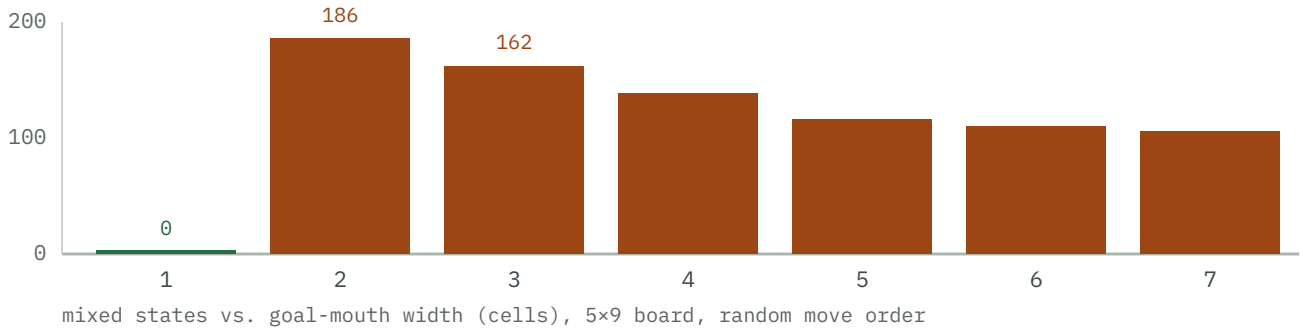
RANDOM MOVE ORDER, HYBRID SOLVER — SAME FIXED POINT, |DIFF| < 3E−9

	ROUNDS / SWEEPS	MATRIX-GAME SOLVES	WALL CLOCK
value iteration	108	9 672	13.8 s
policy iteration	21	1 879	17.2 s

5× fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to < 1e−8. The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here.

RQ3 — Mechanism

A stage game needs mixed strategies only when the goal mouth is more than one cell wide.



The gate. One-cell goal → the carrier has one threat, the defender one response, best responses stay pure. Two or more → the defender must guess. Every height-3 board (width 3–11) has zero mixed states.

Geometry predicts the label (`scripts/geometry_model.py`): a depth-4 decision tree on carrier-frame features separates `mixed` from the rest with **precision 0.96, recall 0.91**. The dominant feature is `defender_can_intercept` — the defender within one move of the carrier's forward cell (`P(mixed) = 0.44` vs `0.002`).

THE 94 MIXED STATES, BY EQUILIBRIUM SUPPORT × DEFENDER GEOMETRY

SUPPORT	DEFENDER	COUNT
2×2 (matching pennies)	ahead of the carrier	40
2×2	directly ahead, same row	24
2×1 / 1×2	directly ahead, same row	12
3×3 / 2×3 / 3×2	ahead	12
2×2	merely adjacent	4
2×1 / 1×2	ahead	2

~88 of 94 have the defender ahead; 68 are a 2×2 matching-pennies mix of carrier {advance, hold} against defender {block, intercept}. All 94 share one value across every equilibrium; 64 have a unique one.

RQ4 — Numerical robustness

The value is three numbers

`value_bracket` returns what the row player guarantees ($\min_k (pM)_k$), gets ($p^\top Mq$), and can reach ($\max_i (Mq)_i$). The true value is bracketed, so the width certifies the LP error.

FINDING

With `scipy`'s HiGHS the three agree to `1.1e-16` per stage game, `8.7e-10` accumulated. The concern is real for older vertex/simplex solvers — a non-issue here.

Discounting and tolerance

	DETERMINISTIC	RANDOM (7×5)
mixed states, γ = 0.5 → 0.9 → 0.995	0 → 0 → 0	122 → 94 → 102
classification split, rel_tol 1e−12 → 1e−3	stable	604 / 1682 / 94 (stable)
classification, rel_tol 1e−2 / 1e−1	—	80 / 0 mixed
fixed 0.1-rounding flips a saddle	0	62

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real `minimax − maximin` gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry mixed region.

Limitations

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Qualitative results hold across 40+ board configurations; the exact `3.9%` is the 7×5 / 3-cell-goal case.
- Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See `docs/assumptions.md`.
- **Claim A** (every converged stage game has a pure saddle) and **Claim B** (a stationary pure equilibrium exists) are established by enumeration. **Claim C** (the theoretical game necessarily has one, over all settings) is *not* — that needs a proof.
- `random` and `coinflip` are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.

Secondary validation — the policy plays correctly

- Duality gap $V0\_br(s_0) + V1\_br(s_0) < 1e{-}9$ for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; `+0.157` empirical discounted return (vs. `+0.150` computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability `0.43` — worse than random — which is the A10 competition's warning about non-equilibrium submissions.

Open questions

- **General-sum.** `markov_game.py` solves 2-player general-sum Markov games by enumerating stage equilibria and selecting one (the notes' "largest sum of values" — a no-op for zero-sum, decisive for Battle of the Sexes). The discretised dog / debate game is next.
- **Function approximation.** A network fit to the stage matrices (`nash_dqn.py`, 5→96→96→16 with biases, 500 epochs) trails the exact solver badly:

	EXACT HYBRID NASH-Q	NEURAL NASH-Q
wall clock	9 sweeps, 0.45 s	500 epochs, 55 s
max |V − V_exact|	0	0.54
action agreement	100%	53%
exploitability	0	0.43

The network fits the ~1 000 zero-value states but misses the γ^k structure — about as exploitable as random play. Keep the exact solver as ground truth.

- **A proof.** Can "one-cell goal ⇒ pure everywhere" be turned into a theorem about the transition structure?

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soccer_nash — game.py · matrix_games.py · support_enum.py · nash_q.py · markov_game.py · numerics.py · geometry.py · symmetry.py · tree.py · best_response.py · exploit.py · mlp.py · nash_dqn.py · opponents.py · shaping.py · simulate.py · experiment.py
scripts — experiments.py · analyze.py · numerics.py · equilibria.py · geometry_model.py · policy_iteration.py · selfplay.py · nash_dqn.py · a10_part1.py · a10_part2.py · a10_competition.py
experiments — baseline.csv · gamma_sweep.csv · tolerance_sweep.csv · board_sweep.csv · goal_mouth_sweep.csv
env: 7×5 grid, 2 380 states, zero-sum ±1, $\gamma = 0.9$